UNIT-1 ASSIGNMENT

SHAIK SHOUIB AKTHAR

CSA5117

192312208

CRYPTOGRAPHY & NETWORK SECURITY

(1)

1. Encrypt the message "DEFEND THE WALL" using Caeser Cipher with key=3.
2. Decrypt the message "KHOOR ZRVOG" using Caeser cipher with key=3.
3. A Caeser Cipher is used to encrypt the word "SECRET" to "VHFUHW". Determine the keyword & decrypt "WKLVLVD WHVW"

96/100

1. Encrypt the DEFEND THE WALL Using key = 3

Rule: shift each letter 3 position forward

Plain Text → D E F E N D   THE → WKH

Cipher Text → G H I H Q G   WALL → ZDOO

Cipher Text : GHIHQG WKH ZDOO

2. Decrypt "KHOOR ZRVOG" with key = 3

Rule: Every letter 3 places backward

K → H   Z → W

H → E   R → O

O → L   V → R

O → L   O → L

R → O   G → D

Cipher Text: HELLO WORLD

3) "SECRET" → "VHFUHW"

Key : S → V = +3 , E → H = +3

Encrypt WLV LVD WHVW.

Shift backward by 3

W → T

K → H

L → I

V → S

LV → IS

D → A

WHVW → TEST.

Cipher : THIS IS A TEST

(2) a) Using the Substitution key

A → Q , B → W, C → E, D → R, E → T, F → Y ... - Z → M

Encrypt the word "FIRE"

b) Given the Cipher text "UJHHD QEB NRFZH YOLTK", and knowing it is encrypted with a monoalphabetic substitution. Find the plaintext using Frequency analysis

a) Key:

A → Q , B → W , C → E , D → R , E → T , F → Y ... - Z → M

Encrypt "FIRE"

F → Y , I → O , R → K , E → T   Cipher Text : YOKT.

b) Frequency Analysis:

Cipher Text :

"UJHHD QEB NRFZH YOLTK"

This is the famous Substitution example

Plain Text :

"THREE THE QUICK BROWN."

Explaination:

Frequency analysis identifies the most common letter and matches them with common English letter like E, T, A, ~~O~~ until meaningful word appear.

(3) a) Encrypt the message "ATTACK AT PALACE" using the keyword "LEMON".

b) Decrypt the cipher text "LXFOPVEFRNHD" using the keyword "LEMON."

c) Given the encrypted message "QNXEPKMAEGTU" using the vignere cipher and keyword "key". find original message

a) keyword = LEMON

Encrypt "ATTACK AT PALACE"

Repeat key LEMON LEMON LEMO

using vignere cipher

Cipher text : LXFOPVEFRNH RMH.

b) Decrypt LXFOPVEFRNHR

key word = LEMON.

plain Text : ATTACKATDAWN.

c) Decrypt "QNXEPKMAEGTU"

keyword = KEY

Repeated key:

KEYKEYKEYKEY

After subsracting key values

Original message : CRYPTOGRAPHY

(4) Given a 2x2 key matrix

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

a) Encrypt the message "HI" using the Hill Cipher.

(use $A=0, B=1 \ldots Z=25$)

b) If the message "PQ" was encrypted to "QC" using Hill Cipher matrix, Verify the result by showing matrix multiplication.

a) Matrix

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

Encrypt HI

$A=0$
$H=7$
$I=8$

Vector $\begin{bmatrix} 7 \\ 8 \end{bmatrix}$

Multiply $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 45 \\ 54 \end{bmatrix}$

Modulo 26

$45 \bmod 26 = 19$
$54 \bmod 26 = 2$

$19 = T, \ 2 = C$

Answer : TC

b) Verify "PQ" → "QC"

$P = 15, \ Q = 16$

Multiply,

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 16 \end{bmatrix} = \begin{bmatrix} 93 \\ 110 \end{bmatrix}$$

Mod 26

$93 \bmod 26 = 15 = P$

$110 \bmod 26 = 6 = G$

∴ PQ does not Encrypt to QC using this matrix.

a) Encrypt the message "MEETMEAFTERLUNCH" using the keyword "ZEBRAS".

b) Decrypt the message "EANMTEFWHETREL" using the keyword "BOARD".

a) plaintext : MEETMEAFTERLUNCH

keyword : ZEBRAS

Repeated:

ZEBRASZEBRASZEB

using vignere Encryption

Cipher Text:

LIFKRWEDXXVCRYGG

b) Cipher text : EANMTEFWHETREL

keyword : BOARD

plaintext : SECRETMESSAGE

⑥ a) Using the one-time pad key XMCKL, Encrypt the message "HELLO".

b) Explain why the message "ATTACKNOW" encrypted using OTP with key "QWERTYUIO" is unbreakable under ideal Condition.

a) key:

XMCKL

plaintext : HELLO

Convert to numbers

H=7, E=4, L=11, L=11, O=14

key:

X = 23, M = 12, C = 2, K = 10, L = 11

Add mod 26

7 + 23 = 30 → 4 = E

4 + 12 = 16 → Q

11 + 2 = 13 → N

11 + 10 = 21 → V

14 + 11 = 25 → Z

Cipher text = EQNVZ.

b)

* The key is completely random
* The key length equal to the message length.
* The key is used only once.
* The key is kept secret.

(7). a) Encrypt the message

PT : WE ARE DISCOVERED

key : ZEBRA

b) Decrypt the message

CT : EVLNEA CDTKESEAQROFDJDEECOXWN

key : TRIAL

c) The message "INFORMATION SECURITY" is encrypted using Columnar transposition with key "HACK."

a) PT = WE ARE DISCOVERED

Remove Space:

WEAREDISCOVERED

key : ZEBRA

You are given the following Cipher text encrypted text using a monoalphabetic Substitution Cipher UZQSOVUOHXMOPVGPV GPOZPEVSGZWSZOPFPESXUZUETUCG. Assume it is an english sentence.

Cipher text

UZQSOVUOHXMOPVGPOZPEVSGZWS ZOF PESXUZU ETUGG

letter Freq

Letter	Count
Z	7
U	5
O	5
P	4
S	4
V	3
E	3

Problem Mapping:

Since E is the most Common letter in English

Z→E U→T O→A P→O S→I

(9) You intrepeted a message encrypted with a monoalphabetiz Substitution Cipher.

CT: BJJWQJW YMJW VXFXNROQJWX

Original plaintext start with "THIS IS THE".

Known plaintext: THIS IS THE

Mapping:

T→B H→J I→W S→Q E→M

Continue Decryption

·THIS IS THE ?? SIMPLE?

Plain Text:

THIS IS THE SIMPLEST

1) Arrange the TEXT

Z E B R A

W E A R E

D I S C O

V E R E D

2) Alphabetic order of keyword

ABERZ

original position

A → 5 B → 3 , E → 2 , R → 4 , Z → 1

3) Read Column in Sorted Order

A : E O D → EOD

B : ASR → ASR

E : EIE → EIE

R : RCE → RCE

Z : WDV → WDV

Cipher text : EODASREIE RCE WDV

b) Ciphertext : EKNEACDTK ESEAQ,ROFOJD EEWXWN.

keyword : TRIAL

WE ARE DISCOVERED FLEE AT ONCE

c) keyword : HACK

L=19 Row needed=5

C=4 Padding required = 1x.

A → NMOCT C → FANWY H → RIEI K → OTSRX

Final Ciphertext

NMOCTFANUYIREIOTSRX.

(10) You are given the cipher text below, encrypted using the vignere cipher with an unknown short key (2-5 letter length):

CT: ZICVTWQNGRZGVTWAVZHCQYGLMCJ

CT: ZICVTWQNGRZGVTWAVZHCQYGLMCJ.

S-1: key length

Repeated Pattern Suggest a 3-letter key.

S-2: Recover key

frequency analysis

S-3: Decrypt:

THISISAVIGENERECIPHERMESSAGE

Explaination:

* Recover the key using frequency analysis.
* Decrypt each letter by Subtracting the key value. mod 26.